A Tensor-based Granular Ball Model for Anomaly Detection

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Abstract—Tensor-based multi-view clustering transforms multi-view data from the object level to the clustering level, whereas anomaly detection requires a continuous ranking of anomaly scores for individual original samples. This difference in the granularity of data representation makes it challenging to directly integrate the two. Replacing samples with granular-ball centers can reduce the quantity of processing units; however, without explicit member decoding, ball-level structural evidence still cannot be directly associated with the original records. To resolve this limitation, this study puts forward a Tensor-based Granular Ball Model for Anomaly Detection (TGBAD), which applies the tensor graph learner to granular-ball centers, preserves the membership relationships between samples and granular balls, and maps the ball-level reconstruction residuals back to individual samples. Subsequently, the multi-tensor granular-ball score, feature-tail rarity, and Isolation Forest scores are fused in a label-free manner using average rank percentiles and a fixed Top-2 rule. On 16 public datasets, TGBAD significantly outperforms 9 of the 10 baselines in terms of AUC; scalability experiments further validate the efficiency of granular execution. The code is publicly available online at https://github.com/from-china-to/TGBAD.

Key Words—Granular ball computing, anomaly detection, tensor graph learning, membership decoding, unsupervised fusion

I. Introduction

Anomaly detection aims to assign higher scores to rare observations deviating from prevailing patterns, and its core output is a continuous ranking at the level of the original records [1], [2] . In an unsupervised setting, models cannot rely on labels to learn component weights or select per-dataset scoring rules; meanwhile, marginal rarity, local isolation, and higher-order structural deviation provide complementary signals from different perspectives. Preserving these sources of evidence under a unified protocol while keeping the final scores traceable to individual samples remains a common challenge in both method design and experimental interpretation.

Tensor-based multi-view clustering jointly represents view-specific and view-shared structures using higher-order variables [3], [4] . Embedded Multi-View Clustering via Fusing Tensor Subspace Representation and Adaptive Graph Learning (EMCTGL) incorporates tensor subspace representation, graph learning, and embedded clustering into one backbone [5] , and anchor-based tensor methods further reduce the computational scale with fewer structural units [6] . However, these methods target clustering representation: for N samples and V views, maintaining an $N \times N \times V$ tensor variable directly limits the scale, and center-/clustering-level outputs cannot be directly converted into a continuous sample-level anomaly ranking.

Granular balls compactly represent sample groups through center, size, radius, and member set, providing a natural interface for cross-scale representation [7], [8] . Granular balls have been applied to anomaly detection, e.g., fuzzy granular random walks, multi-scale granular-ball density, and fuzzy neighborhood entropy [9]–[11] . However, when the inherited tensor graph learner runs on granular-ball centers, decoding ball-level structural evidence back to individual samples via explicit membership and fusing it with sample-level evidence without labels remain open problems.

To address these challenges, we propose TGBAD: it constructs granular balls in the joint space of normalized feature blocks, runs the inherited EMCTGL learner on per-view granular-ball centers, and decodes ball-level reconstruction residuals to individual samples via member assignment. To compensate for within-ball sharing of structural scores, two-sided feature-tail rarity and Isolation Forest (IF) [12] scores are further computed at the sample level; after normalizing all components by average rank percentiles, the two largest per sample are averaged as the final anomaly score.

We summarize the key contributions of our study as follows:

- We propose a granular-ball execution–sample decoding interface with preserved membership relationships, which explicitly maps granular-ball-level tensor reconstruction residuals to structural anomaly scores at the scale of the original records;
- We develop a fixed, label-free percentile-based Top-2 scoring protocol;
- We conduct experiments on 16 public datasets against 10 comparison algorithms to evaluate the performance and scalability of the proposed method.

The structure of the remaining sections is outlined below: Section 2 details the proposed TGBAD; Section

3 reports the experiments and analyses; and Section 4 concludes the paper.

II. Proposed Approach

A. Feature Block View and Granular-ball Execution

For a dataset with N objects and D attributes, let the input table be $X \in \mathbb{R}^{N\times D}$. Under the fixed protocol, the attributes are approximately divided into $V = \min(3, D)$ feature blocks $X^{(v)} \in \mathbb{R}^{N\times D_v}$ according to their original feature order. Each view first undergoes feature-wise Min-Max scaling and sample-wise ℓ_2 normalization, after which the views are concatenated to form the joint representation z_i. If the joint dimensionality exceeds 128, a randomized SVD with random seed 0 projects the joint representation to 128 dimensions.

A granular ball is a compact information granule characterized by its center, radius, and member set, and serves as the basic processing unit of the proposed method [7], [8] . Let the minimum granular ball size be $n'_{\min} = \min\{8, \max(1, \lfloor N/2 \rfloor)\}$, following the minimum capacity threshold of granular-ball-based multi-view learning [13] . The maximum number of granular balls is defined as $M_{\max} = \min\{N, 256, \max(2, \lfloor N/n'_{\min} \rfloor)\}$. The algorithm starts from a single set containing all samples and considers only balls with size at least $2n'_{\min}$ as candidates. Following a dispersion-first strategy, it preferentially splits the candidate with the largest product $\rho_m\sqrt{n_m}$ using the 2-Means method [7] , where the radius ρ_m of the m-th granular ball with member set $\mathcal{I}_m$ and size n_m is defined as

$$\rho_m^2 = \frac{1}{n_m V} \sum_{i\in\mathcal{I}_m} \left\| z_i - \frac{1}{n_m} \sum_{j\in\mathcal{I}_m} z_j \right\|^2, \tag{1}$$

in which z_i denotes the coordinate of sample i in the joint representation space. If the 2-Means split violates the minimum-size constraint, the samples are stably sorted along the dimension with the largest variance and then divided into two balanced groups. Suppose that the final partition consists of M granular balls. The m-th granular ball stores its member set $\mathcal{I}_m$, size n_m, radius, and membership assignment $a_i = m$. Its center in the v-th view is given by

$$c_m^{(v)} = \frac{1}{n_m} \sum_{i\in\mathcal{I}_m} x_i^{(v)}. \tag{2}$$

The same membership assignment is used across all views, thereby preserving a deterministic mapping between the center matrix $C^{(v)} \in \mathbb{R}^{M\times D_v}$ and the original samples. The granular-ball mode uses the M centers as model units, whereas the full-sample mode corresponds to $M = N$ and $a_i = i$.

B. Center-Level Tensor Learning and Membership Decoding

The centers from individual views are embedded into an expanded data tensor $\mathcal{X} \in \mathbb{R}^{D_\Sigma\times M\times V}$, where $D_\Sigma = \sum_v D_v$. Let $\mathcal{B} \in \mathbb{R}^{M\times M\times V}$ denote the learned self-representation tensor, obtained by the EMCTGL solver [5] , whose objective jointly optimizes a TLog-induced tensor low-rank term, a tensor subspace reconstruction term, view-weight smoothness, affinity graph learning with sparsity, and embedded clustering, with all hyperparameters kept at their defaults. The ball-level reconstruction residual is defined as

$$r_m^{\mathrm{rec}} = \left\| \mathcal{X}_{:,m,:} - (\mathcal{X} * \mathcal{B})_{:,m,:} \right\|_F, \tag{3}$$

where $*$ denotes the t-product [14] , which performs matrix multiplication between corresponding frontal slices in the discrete Fourier transform (DFT) domain, i.e., $\widehat{\mathcal{X} * \mathcal{B}}^{(i)} = \widehat{\mathcal{X}}^{(i)}\widehat{\mathcal{B}}^{(i)}$ for $i = 1, \ldots, V$, followed by an inverse DFT along the third mode. Sample-level decoding is determined by the membership assignment:

$$r_i^{\mathrm{rec}} = r_{a_i}^{\mathrm{rec}}. \tag{4}$$

Eq. (4) allows the structural score of each sample to be traced to its corresponding granular ball. Let the membership matrix $P \in \{0, 1\}^{N\times M}$ satisfy $P_{im} = \mathbb{I}(a_i = m)$, and let the ball-level residual vector be $\bar{r}^{\mathrm{rec}} = [r_1^{\mathrm{rec}}, \ldots, r_M^{\mathrm{rec}}]^\top$. Then, Eq. (4) is equivalent to $r^{\mathrm{rec}} = P\bar{r}^{\mathrm{rec}}$. Since each row of P contains exactly one nonzero entry, the decoding result requires neither additional nearest-neighbor searches nor interpolation parameters, and satisfies

$$\frac{1}{N}\mathbf{1}_N^\top r^{\mathrm{rec}} = \frac{1}{N}\sum_{m=1}^{M} n_m r_m^{\mathrm{rec}}, \tag{5}$$

that is, the mean sample-level structural score equals the granular-ball-size-weighted mean of the ball-level scores.

Since a single residual signal may be insufficient, a second ball-level source of evidence is introduced, which is radius density:

$$d_m = \rho_m \sqrt{\frac{\tilde{n}}{n_m}}, \qquad \tilde{n} = \mathrm{median}(\{n_1, \ldots, n_M\}), \tag{6}$$

which assigns higher values to granular balls that are small in size and highly dispersed, making them more suspicious. Both the reconstruction residual and radius density are defined over the M granular balls. Let $\mathrm{pr}_M(\cdot)$ denote the average rank percentile that maps a vector of length M to $[0, 1]$, as defined in Eq. (13). The ball-level structural score is then obtained by taking the maximum of the two normalized components for each granular ball:

$$b_m = \max\left\{\mathrm{pr}_M(r^{\mathrm{rec}})_m,\ \mathrm{pr}_M(d)_m\right\}, \tag{7}$$

and the decoded sample-level structural score is b_{a_i}.

C. Multi-Tensor Granular-Ball Scores and Label-Free Fusion

To compensate for the limitation that samples within the same granular ball share the same ball-level structural score, the method first computes the within-ball deviation.

The distance between the i-th sample and the center of its assigned granular ball in the joint space is defined as

$$e_i = \left(\frac{1}{V} \sum_{v=1}^{V} \left\| x_i^{(v)} - c_{a_i}^{(v)} \right\|^2 \right)^{1/2}, \tag{8}$$

where $x_i^{(v)}$ denotes the preprocessed representation of sample i in the v-th view. The multi-tensor granular-ball score is obtained by averaging the ball-level structural score and the average rank percentile of the within-ball deviation:

$$g_i = \frac{1}{2} \left(b_{a_i} + \mathrm{pr}_N(e)_i \right), \tag{9}$$

where b_{a_i} is already in the percentile space $[0, 1]$, while $\mathrm{pr}_N(e)_i$ is the average rank percentile of e_i computed over all samples; the two components are therefore on the same scale. Equation (9) simultaneously preserves the inter-ball structural evidence (b_{a_i}) and the within-ball resolution (e_i).

In addition, the method computes two sample-level supporting components. For feature d, let q_{id} denote the average rank percentile of x_{id}, computed by Eq. (13) applied to the d-th feature column. Its two-sided tail rarity is defined as

$$t_{id} = -\log \left[\max \left(2 \min(q_{id}, 1 - q_{id}), \frac{1}{N} \right) \right]. \tag{10}$$

When the number of distinct values of a feature does not exceed $\min\{32, \max(2, \lfloor \sqrt{N} \rfloor)\}$, the empirical frequency of each value is used in place of the two-sided tail quantile $2\min(q_{id}, 1 - q_{id})$ in Eq. (10). Under the fixed protocol, the rarity scores are first averaged over all features within each view and then averaged across views to obtain r_i^{tail}.

Let $\mathcal{D}_v$ denote the feature index set of the v-th view, with $D_v = |\mathcal{D}_v|$. In the main experiments, all features are included in the aggregation, yielding

$$r_i^{\mathrm{tail}} = \frac{1}{V} \sum_{v=1}^{V} \frac{1}{D_v} \sum_{d \in \mathcal{D}_v} t_{id}. \tag{11}$$

The input to Isolation Forest is first subjected to view-wise Min-Max scaling and divided by $\sqrt{D_v}$, followed by concatenation. Let $\hat{x}_i^{(v)}$ denote the scaled sample in the v-th view. The balanced joint vector and anomaly score are defined as

$$\tilde{x}_i = \left[\frac{\hat{x}_i^{(1)}}{\sqrt{D_1}}; \ldots; \frac{\hat{x}_i^{(V)}}{\sqrt{D_V}} \right], \quad r_i^{\mathrm{iso}} = 2^{-\mathbb{E}[h(\tilde{x}_i)]/c(\psi)}, \tag{12}$$

where $h(\tilde{x}_i)$ denotes the path length of sample $\tilde{x}_i$ in an isolation tree, $\psi = \min(256, N)$, $c(\psi) = 2H_{\psi-1} - 2(\psi-1)/\psi$, $\mathbb{E}[\cdot]$ denotes the expectation over the ensemble of isolation trees, and H denotes the harmonic number, estimated by $\ln(\cdot) + 0.5772$ [12] .

Algorithm 1 Membership-Preserving Execution and Sample-Wise Decoding of TGBAD

Input: Data matrix $X \in \mathbb{R}^{N \times D}$, view number V, granular-ball bounds $M_{\max}$ and $n'_{\min}$
Output: Sample-level anomaly scores $s \in [0, 1]^N$

1: Construct V block views by feature order with Min-Max and sample-wise ℓ_2 normalization, and concatenate them into the joint space
2: Recursively bisect the joint space into granular balls by dispersion-first splitting (radius $\times\sqrt{\text{size}}$, Eq. (1)); store $\{\mathcal{I}_m, n_m, a_i\}$
3: Compute per-view centers $C^{(v)}$ (Eq. (2)), embed them into tensor $\mathcal{X}$, and run the inherited EMCTGL solver to obtain $\mathcal{B}$
4: Compute ball-level structural scores b_m via Eqs. (3)–(7)
5: Decode b_{a_i} to samples and fuse with within-ball deviations to obtain multi-tensor granular-ball scores g_i via Eqs. (8)–(9)
6: Compute sample-level scores r_i^{tail} (Eqs. (10)–(11)) and r_i^{iso} (Eq. (12))
7: Normalize $\{r^{(\mathrm{gbs})}, r^{\mathrm{tail}}, r^{\mathrm{iso}}\}$ to average rank percentiles $p^{(k)}$ via Eq. (13)
8: return s_i as the mean of the two largest percentiles via Eq. (14)

The three components are on different scales. For $k \in \{\mathrm{gbs}, \mathrm{tail}, \mathrm{iso}\}$, the average rank percentile

$$p_i^{(k)} = \frac{\mathrm{rank}_{\mathrm{avg}}(r_i^{(k)}) - 1}{N - 1} \tag{13}$$

is used to transform the corresponding scores to $[0, 1]$. The final score is defined as the mean of the two largest percentiles for each sample:

$$s_i = \frac{1}{2} \sum_{p \in \mathrm{Top2}\{p_i^{(\mathrm{gbs})}, p_i^{(\mathrm{tail})}, p_i^{(\mathrm{iso})}\}} p. \tag{14}$$

III. Experiments

We conduct extensive experiments comparing TGBAD with existing algorithms. The experimental setup is described first, followed by a comparison and analysis of the experimental results. We perform hypothesis testing to evaluate the statistical significance of the proposed method, and we also conduct scalability experiments.

A. Experimental Setup

The effectiveness of TGBAD is evaluated on 16 datasets from the public UCI database. The datasets are obtained from publicly available repositories[1,2]. The datasets with their abbreviations are as follows: Abalone_variant1 (Abalone), Annthyroid (AnnThy), Bands_band_6_

[1]https://github.com/from-china-to/Outlier-detection
[2]https://github.com/BELLoney/Outlier-detection

variant1 (Bands6), Chess_nowin_34_variant1 (Chess34), Glass (Glass), Hepatitis_2_9_variant1 (Hepatitis), Letter (Letter), Lymphography (Lymph), Mammography (Mammo), Monks_0_4_variant1 (Monk), Mushroom_p_85_variant1 (Mushroom), Pageblocks_1_258_variant1 (Page), Spambase_spam_56_variant1 (Spam), Thyroid (Thyroid), Tic_tac_toe_negative_12_variant1 (Tic12), Zoo_variant1 (Zoo).

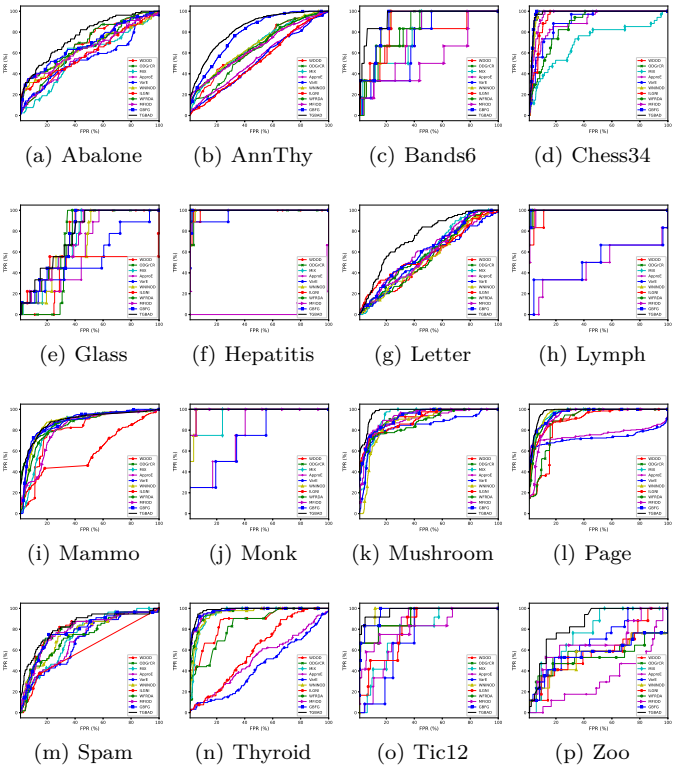
For the selected datasets, the performance of TGBAD is compared with the following 10 algorithms.

1) Weighted density-based outlier detection algorithm (WDOD) (2014) [15]
2) Outlier detection algorithm based on granular computing and rough sets (ODGrCR) (2015) [16]
3) Joint learning-based outlier detection algorithm for mixed data (MIX) (2019) [17]
4) Approximate structure score-based outlier detection algorithm (ApproE) (2020) [18]
5) Variance structure score-based outlier detection algorithm (VarE) (2020) [18]
6) Weighted neighborhood information network-based outlier detection algorithm (WNINOD) (2021) [19]
7) Local–global neighborhood information-based outlier detection algorithm (ILGNI) (2023) [20]
8) Weighted fuzzy–rough density-based outlier detection algorithm (WFRDA) (2023) [21]
9) Multi-scale fuzzy information fusion-based outlier detection algorithm (MFIOD) (2024) [22]
10) Granular-ball fuzzy information-based outlier detection algorithm (GBFG) (2025) [23]

The baselines follow their original protocols: continuous attributes of WDOD and ODGrCR are discretized by Fuzzy C-Means; the regularization parameters of ApproE and VarE are searched over $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$; ILGNI uses $a = 1, b = 100$; the fuzzy similarity scale of WFRDA is searched over $\{1, 2, \ldots, 20\}$; the scale parameter of MFIOD is searched over $(0, 1)$ with step 0.001; MIX, WNINOD, and GBFG use their publicly reported best settings. For each dataset, the best-parameter original scores of each baseline are retained.

The fixed configuration of TGBAD uses a ball-count cap of 256, a minimum ball size of 8, at most 128 ball features, an embedding dimension of 2, and 15 graph neighbors; splitting prioritizes the largest radius$\times\sqrt{\text{size}}$ product. The inherited solver uses $\lambda = 10$, $\lambda_1 = 10^{-2}$, $\lambda_2 = 0.1$, $\beta = 1$, and $\sigma = 10$, with convergence threshold 10^{-3} and at most 50 iterations; Isolation Forest uses 100 trees and a subsample size of at most 256. All random seeds are 0.

The metrics are ROC curves, AUC, and F1-SCORE (the maximum over all thresholds, used only for offline evaluation). In the scalability experiments, the granular-ball and full-sample modes share the same feature partitioning, solver, scoring rules, and random state, and are run alternately with up to three repetitions (600 s per run); runtimes are reported as medians of completed runs and AUCs as their means.



(a) Abalone (b) AnnThy (c) Bands6 (d) Chess34

(e) Glass (f) Hepatitis (g) Letter (h) Lymph

(i) Mammo (j) Monk (k) Mushroom (l) Page

(m) Spam (n) Thyroid (o) Tic12 (p) Zoo

Figure 1. Experimental comparison results on ROC.

TABLE 1
EXPERIMENTAL EVALUATION OF AUC.

Dataset	TGBAD	WDOD	ODGrCR	MIX	ApproE	VarE	WNINOD	ILGNI	WFRDA	MFIOD	GBFG
Abalone	0.7610	0.6492	0.6599	0.5717	0.5930	0.5480	0.6410	0.6040	0.6614	0.6422	0.6806
AnnThy	0.8250	0.6548	0.6247	0.6611	0.5470	0.5226	0.6671	0.5223	0.6579	0.6461	0.7834
Bands6	0.9402	0.7345	0.8237	0.7719	0.7425	0.7025	0.8825	0.8745	0.8232	0.5454	0.8942
Chess34	0.9708	0.9654	0.9662	0.6996	0.8880	0.8920	0.9644	0.9346	0.8625	0.9565	0.9692
Glass	0.7591	0.6791	0.6778	0.7176	0.6287	0.5382	0.6743	0.4818	0.7458	0.7252	0.7420
Hepatitis	1.0000	0.9856	0.9882	0.9987	0.0327	0.9621	0.9922	0.9948	0.9948	0.9908	0.9974
Letter	0.7383	0.5294	0.5382	0.5795	0.5938	0.5292	0.5374	0.5816	0.5844	0.5740	0.6072
Lymph	0.9988	0.9742	0.9953	0.9977	0.4765	0.5070	0.9918	0.9906	0.9930	0.9859	0.9965
Mammo	0.8838	0.7936	0.8640	0.8322	0.8675	0.8659	0.8681	0.5871	0.8476	0.8102	0.8730
Monk	1.0000	0.9868	1.0000	0.9397	0.7719	0.7270	0.9781	1.0000	1.0000	0.9868	1.0000
Mushroom	0.9712	0.9118	0.9240	0.9223	0.9008	0.8304	0.8664	0.8979	0.8734	0.9211	0.9218
Page	0.9683	0.8663	0.8814	0.9560	0.7600	0.7269	0.9560	0.9145	0.9440	0.9220	0.9557
Spam	0.8266	0.7935	0.7899	0.7304	0.6979	0.6847	0.7543	0.6073	0.7249	0.7822	0.7806
Thyroid	0.9775	0.8704	0.8293	0.9406	0.5076	0.4316	0.9427	0.5957	0.9568	0.9687	0.9730
Tic12	0.9752	0.9076	0.9121	0.7714	0.7983	0.7330	0.9667	0.8126	0.9135	0.8244	0.9615
Zoo	0.8382	0.5413	0.5707	0.7689	0.3235	0.6828	0.5385	0.5889	0.5259	0.6723	0.5630
Average	0.9021	0.8027	0.8153	0.8037	0.6331	0.6802	0.8263	0.7493	0.8193	0.8096	0.8562
1st Order	16	0	1	0	0	0	0	1	1	0	1

B. Experimental Results

1) ROC Curves: Figure 1 summarizes the ROC curves of the 11 algorithms across the 16 datasets, with the black curve representing TGBAD. On Lymph, Monk, Mushroom, Tic12, Page, Thyroid, and Hepatitis, TGBAD reaches high TPRs at relatively low FPRs, with its curves generally closer to the upper-left corner; on Zoo, the multi-tensor granular-ball score also yields a ROC curve that is clearly above those of the other 10 algorithms.

2) AUC Scores: Table 1 shows that TGBAD attains the highest (or tied-highest) AUC on all 16 datasets (mean 0.9021, +0.0459 over GBFG), reaching 1.0000 on Monk

TABLE 2
EXPERIMENTAL EVALUATION OF F1-SCORE.

Dataset	EMTGBAD	WDOD	ODGrCR	MIX	ApproE	VarE	WNINOD	ILGNI	WFRDA	MFIOD	GBFG
Abalone	0.1905	0.0612	0.0640	0.0660	0.0674	0.0797	0.1905	0.2000	0.1654	0.1660	0.1789
AnnThy	0.3490	0.2366	0.1919	0.2212	0.1476	0.1459	0.2453	0.1473	0.2439	0.2765	0.2747
Bands	0.3529	0.1667	0.2222	0.2222	0.1111	0.1250	0.2500	0.3636	0.1818	0.2857	0.3077
Chess	0.4526	0.4317	0.4247	0.1287	0.2400	0.2585	0.3973	0.2755	0.1818	0.3368	0.4559
Glass	0.1818	0.1800	0.2022	0.1651	0.1379	0.1429	0.1471	0.1739	0.1875	0.1667	0.1782
Hepatitis	1.0000	0.8421	0.8571	0.9474	0.1748	0.8421	0.8889	0.8889	0.9000	0.9474	0.9412
Letter	0.2465	0.1321	0.1353	0.1466	0.1571	0.1298	0.1312	0.1784	0.1490	0.1407	0.1592
Lymphography	0.9231	0.8000	0.8333	0.9231	0.1739	0.2857	0.8333	0.8333	0.8333	0.7692	0.9091
Mammography	0.3317	0.1581	0.3745	0.1664	0.2530	0.2544	0.2254	0.1234	0.1914	0.1544	0.2692
Monks	1.0000	0.8571	1.0000	0.8571	0.3333	0.3333	0.6667	1.0000	1.0000	0.8571	1.0000
Mushroom	0.5789	0.5414	0.4685	0.2832	0.4096	0.2365	0.2000	0.5254	0.5077	0.3245	0.4103
PageBlocks	0.5802	0.3398	0.3508	0.5663	0.6324	0.5960	0.5232	0.5306	0.5075	0.4190	0.5828
Spambase	0.2312	0.1513	0.1570	0.0974	0.1105	0.1186	0.1567	0.0909	0.1305	0.1401	0.1385
Thyroid	0.5701	0.5547	0.2986	0.3352	0.0552	0.0482	0.3849	0.0642	0.4486	0.5842	0.6154
Tic12	0.8000	0.8000	0.6364	0.1205	0.1023	0.0914	0.5185	0.2222	0.7619	0.3636	0.5333
Zoo	0.6000	0.3902	0.4390	0.4865	0.2957	0.4000	0.3684	0.3871	0.4138	0.4865	0.4286
Average	0.5243	0.4152	0.4160	0.3583	0.2126	0.2555	0.3830	0.3753	0.4253	0.4012	0.4614
1st Order	9	1	3	1	1	0	0	3	1	0	3

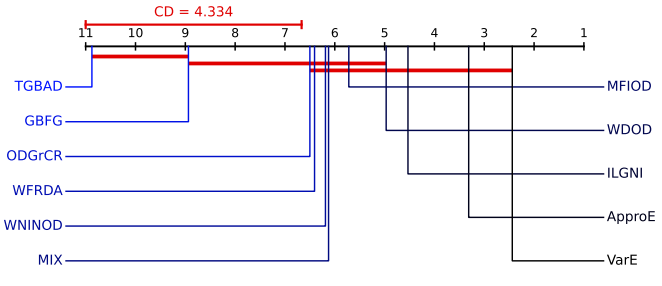


Figure 2. Nemenyi test figure on AUC.

TABLE 3
Execution Cost and Ranking Quality of Granular-Ball vs. Full-Sample Modes.

Dataset	N	M	M/N	N/M	GB(s)	Full(s)	Time ratio	GB AUC	Full AUC	ΔAUC
Abalone	4177	256	0.0613	16.32	7.628	Timeout(>600 s)	–	0.7610	–	–
AnnThy	7200	256	0.0356	28.13	7.269	Timeout(>600 s)	–	0.8250	–	–
Bands6	318	34	0.1069	9.35	0.715	12.682	17.73	0.9402	0.9621	−0.0219
Chess34	1703	161	0.0945	10.58	4.256	329.383	77.39	0.9708	0.9707	+0.0001
Glass	214	18	0.0841	11.89	0.508	5.707	11.24	0.7591	0.7703	−0.0112
Hepatitis	94	10	0.1064	9.40	0.409	1.625	3.97	1.0000	0.9991	+0.0009
Letter	1600	156	0.0975	10.26	3.977	243.134	61.14	0.7383	0.6774	+0.0609
Lymph	148	14	0.0946	10.57	0.464	2.885	6.21	0.9988	0.9984	+0.0004
Mammo	11183	256	0.0229	43.68	7.663	Memory limit	–	0.8838	–	–
Monk	232	23	0.0991	10.09	0.518	3.067	5.92	1.0000	1.0000	0.0000
Mushroom	4293	256	0.0596	16.77	8.522	Timeout(>600 s)	–	0.9712	–	–
Page	5171	256	0.0495	20.20	7.477	Timeout(>600 s)	–	0.9683	–	–
Spam	2844	256	0.0900	11.11	10.729	Timeout(>600 s)	–	0.8266	–	–
Thyroid	3772	256	0.0679	14.73	7.045	Timeout(>600 s)	–	0.9775	–	–
Tic12	638	65	0.1019	9.82	1.202	37.955	31.57	0.9752	0.9698	+0.0054
Zoo	101	9	0.0891	11.22	0.398	1.824	4.58	0.8382	0.7102	+0.1280

and Hepatitis. Even on Zoo (101 samples, 16.83% anomalies), TGBAD still outperforms the strongest baseline (MIX) by 0.0693, confirming that the granular-ball score remains effective on sparse, high-ratio data.

3) F1 Scores: Table 2 shows a mean F1-SCORE of 0.5243 (+0.0629 over GBFG) and the row-best value on 9 datasets; on these datasets, TGBAD attains strong AUC and F1 simultaneously, indicating that its ranking advantage translates into effective anomaly identification.

4) Statistical Hypothesis Testing: On the 11×16 AUC matrix, the Friedman test rejects the null hypothesis of equivalent performance at $\alpha = 0.01$ ($\chi_F^2 = 82.0755$, $p = 1.97 \times 10^{-13}$; Iman–Davenport $F(10, 150) = 15.7990 > 2.4412$, $p = 4.03 \times 10^{-19}$) [24], [25] . In the Nemenyi post-hoc test ($CD_{0.01} = 4.334$, Figure 2), TGBAD attains the top mean rank (10.875 vs. 8.9375 for GBFG) and is not connected to 9 of the 10 baselines; only the difference from GBFG lacks consistent statistical evidence.

5) Scalability Experiments: Table 3 shows that the granular-ball mode finishes all 16 datasets within 11 s using $M \leq 256$ units, whereas the full-sample mode completes within 600 s on only 9 datasets (the remaining six timed out on every repetition, and Mammography was skipped because its estimated peak memory of 78 GB exceeds the 60 GB limit). Both modes run the configuration of the proposed method (dispersion-first splitting, the multi-tensor granular-ball score, and the fixed Top-2 fusion), so the granular-ball AUC column coincides with Table 1. ΔAUC denotes the granular-ball AUC minus the full-sample AUC, where the full-sample mode instantiates the same scoring rule with singleton balls, for which the ball-level structural evidence degenerates. Across the 9 completed datasets, ΔAUC ranges from -0.0219 to $+0.1280$ with a mean of $+0.0181$ and is non-negative on 7 of the 9 datasets; granular execution therefore preserves ranking quality at a substantially lower cost.

IV. Conclusion

This paper proposes TGBAD, which connects center-level tensor learning with sample-level anomaly ranking through shared granular-ball membership: the ball-level structural score (reconstruction residual and radius density) is fused with intra-ball deviation into the multi-tensor granular-ball score, and then combined with feature-tail and isolation scores by a fixed mean-rank Top-2 protocol. On 16 datasets, TGBAD achieves a mean AUC of 0.9021 and a mean F1-SCORE of 0.5243, with the highest AUC on all 16 datasets. Granular execution limits the model units to 256, leaving room for larger datasets. Future work will investigate label-free confidence estimation for granular-ball scores.

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